

KUUSAMO KIUTAKONGAS, Finland

WMO: 028110

Lat: 66.3700N

Lon: 29.3119E

Elev: 541

StdP: 14.41

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.6	-24.2	-36.4	0.7	-29.6	-30.5	1.0	-24.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	78.1	61.6	74.2	59.7	70.5	58.6	64.9	72.8	62.8	70.7	60.7	67.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	84.8	67.7	59.9	78.6	65.4	57.7	72.4	63.7	30.2	73.0	28.6	71.0	27.2	67.6	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)																

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.0	6.8	10.9	18.4	31.0	43.3	51.5	58.8	54.4	45.9	33.5	24.2	16.0
	DBStd	19.91	14.92	15.89	11.40	8.28	8.17	7.34	6.52	6.08	5.97	7.42	10.38	14.11
	HDD50	6816	1338	1095	981	571	241	69	8	25	148	512	775	1053
	HDD65	11701	1803	1515	1446	1021	673	407	210	332	574	977	1225	1518
	CDD50	615	0	0	0	0	33	115	280	162	25	0	0	0
	CDD65	24	0	0	0	0	0	3	17	4	0	0	0	0
	CDH74	353	0	0	0	0	23	55	217	58	0	0	0	0
	CDH80	50	0	0	0	0	1	6	32	11	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	21.8	1.4	1.2	1.1	1.1	1.8	2.5	3.1	2.4	2.0	1.9	1.7	1.5
	PrecMax	28.4	2.8	3.1	2.0	1.7	3.0	5.0	5.6	5.5	3.8	3.6	3.7	2.7
	PrecMin	17.3	0.6	0.2	0.3	0.4	0.8	0.6	1.1	0.9	0.4	0.3	0.4	0.3
	PrecStd	3.1	0.6	0.7	0.5	0.4	0.6	1.1	1.0	1.0	1.0	0.9	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	38.1	43.5	59.5	77.2	80.4	83.6	80.8	67.0	51.4	42.8	36.1
		MCWB	33.8	35.7	36.8	46.2	56.2	60.7	63.7	66.8	57.4	49.5	41.5	34.7
	2%	DB	31.7	34.9	38.3	49.9	71.2	74.1	79.7	73.9	62.4	47.7	39.8	34.4
		MCWB	31.0	33.2	33.6	40.6	54.3	59.5	63.1	60.7	54.6	45.4	38.7	33.3
	5%	DB	29.1	32.9	35.8	46.1	64.6	69.4	75.9	69.8	59.1	45.4	36.9	33.1
		MCWB	28.4	31.7	32.1	38.2	52.7	56.0	60.7	59.9	53.0	43.5	36.1	32.3
	10%	DB	26.2	30.4	33.8	43.0	59.0	65.5	72.2	65.8	56.1	43.2	35.0	31.7
		MCWB	25.5	29.2	31.3	36.2	48.6	54.2	60.0	58.0	51.8	41.7	34.3	31.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	36.0	37.2	46.3	58.3	64.9	68.0	67.1	58.8	50.1	42.2	35.0
		MCDB	34.7	37.7	42.2	58.3	70.4	72.3	77.0	78.0	64.3	50.9	42.6	35.7
	2%	WB	31.0	33.7	34.8	41.4	55.5	61.8	65.6	63.5	56.0	45.8	39.0	33.6
		MCDB	31.5	34.8	37.5	48.6	70.4	70.3	73.3	70.4	60.9	46.9	39.8	34.2
	5%	WB	28.5	31.9	33.1	39.1	52.7	58.7	63.9	61.1	54.0	43.8	36.1	32.4
		MCDB	29.0	33.0	35.1	43.6	63.9	67.1	71.8	66.7	57.7	45.1	36.7	33.1
	10%	WB	25.5	29.2	31.5	37.6	49.3	55.8	62.0	59.2	52.5	41.9	34.4	30.9
		MCDB	26.2	30.3	33.5	42.2	57.3	62.9	70.0	64.8	55.5	43.2	34.9	31.5
Mean Daily Temperature Range	5% DB	MDBR	15.5	15.1	21.7	19.3	19.5	17.8	18.3	17.3	14.5	9.0	9.2	12.1
		MCDBR	14.3	10.1	16.7	22.6	30.3	25.7	25.3	24.0	21.0	10.3	8.2	8.6
		MCWBR	14.3	9.4	13.2	14.2	15.7	11.9	11.3	12.8	13.8	8.8	8.1	8.4
	5% WB	MCDBR	14.4	10.3	12.0	18.7	28.1	21.2	21.1	19.4	17.3	10.2	8.0	8.4
		MCWBR	14.4	9.6	9.8	12.3	15.1	11.2	11.3	11.6	12.0	9.0	7.9	8.3
Clear-Sky Solar Irradiance	taub	0.199	0.265	0.299	0.332	0.346	0.345	0.363	0.341	0.316	0.286	0.352	0.199	
	taud	1.924	2.092	2.140	2.212	2.427	2.472	2.445	2.478	2.531	2.344	2.680	1.948	
	Ebn at Noon	67	182	237	261	271	274	265	259	241	181	72	6	
	Edh at Noon	5	17	28	34	30	30	30	26	20	14	5	1	
All-Sky Solar Radiation	RadAvg	25	186	654	1219	1508	1603	1487	1081	576	188	41	3	
	RadStd	2	34	62	116	139	115	152	108	59	32	6	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page